

Digit-Set Preservation under Reversal Addition

Finite automata, core compression, and exact small columns

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Abstract

For an integer base $n \geq 2$, let $T(n, k)$ denote the number of k -element subsets $S \subseteq \{1, \dots, n-1\}$ for which there exists a positive integer N such that the canonical base- n expansions of both N and $N + \text{Rev}_n(N)$ use exactly the digits of S , with no zero digit. We encode the problem by a finite two-sided carry automaton whose symbolic states record two boundary carries, the input and output digit masks, and the rational affine row space generated by local addition equations. This gives eventual quasipolynomiality of $T(n, k)$ for every fixed k .

The local equations have digit support at most three. We prove that every primitive accepted rank- r core, for $r \geq 3$, contains at most $3r - 2$ essential digits. An exact finite rank-filtered certification excludes eventual accepted systems of rank at most two. It follows that, for every fixed $k \geq 3$,

$$T(n, k) = \Theta(n^{k-3}),$$

and the eventual quasipolynomial has degree exactly $k - 3$.

For rank three, the structural support bound reduces the primitive classification to supports at most seven. Exact certifications exclude new primitive rank-three cores on six and seven digits. The neutral-liftable dominant families therefore already occur on at most five digits. Their aggregate volumes yield, for every fixed $k \geq 4$,

$$T(n, k) = \frac{k^2 + k - 4}{4(k-3)!} n^{k-3} + O(n^{k-4})$$

along odd bases, and

$$T(n, k) = \frac{k(k+1)}{4(k-3)!} n^{k-3} + O(n^{k-4})$$

along even bases. Independently of the fixed-column asymptotics, we prove a dense-set construction: for every $n \geq 4$, every k -subset of nonzero digits is realizable once $k \geq \lceil n/2 \rceil + 1$. Hence the right-hand tail of every row is exactly a tail of Pascal's triangle,

$$T(n, k) = \binom{n-1}{k}.$$

We also give complete eventual formulas for the columns $k \leq 5$, including an exact period 420 for $T(n, 4)$ and an exact rank decomposition for $T(n, 5)$.

Verification status. Theorems 1.2 and 1.3, together with the structural lemmas on neutral layers, rank, and core compression, are mathematical proofs without computational input. Theorems 1.4 and 1.5 are computer-assisted only through the finite exact exclusions

stated in Propositions 6.1 and 9.1 and through the exact rank-three macro classification on at most five digits. The exact classifications in Sections 11–12 use finite symbolic searches with exact rational arithmetic. Appendix B lists the computational claims on which the global theorems depend. No numerical floating-point test is used in those certificates.

1 Definition and main results

Let

$$N = (d_{L-1} \cdots d_1 d_0)_n, \quad d_{L-1} \neq 0,$$

be the canonical base- n expansion of a positive integer N . Define

$$\mathcal{D}_n(N) = \{d_0, \dots, d_{L-1}\}$$

and

$$\text{Rev}_n(N) = (d_0 d_1 \cdots d_{L-1})_n,$$

where the displayed reversed word is interpreted as a base- n integer; leading zeros in the reversed word, if any, have no numerical effect. In the present problem, however, zero will be excluded from both digit sets.

Definition 1.1. A set $S \subseteq \{1, \dots, n-1\}$ is *realizable in base n* if there exists $N > 0$ such that

$$\mathcal{D}_n(N) = \mathcal{D}_n(N + \text{Rev}_n(N)) = S.$$

For $1 \leq k \leq n-1$, put

$$T(n, k) = \#\{S \subseteq \{1, \dots, n-1\} : |S| = k, S \text{ is realizable}\}.$$

The triangle $T(n, k)$ has been assigned OEIS number A398831.

Thus neither the input nor the output expansion contains the digit zero.

At first sight, preservation of a digit set under reversal addition may appear to be a rather local property of base- n expansions. In fact, the condition is strongly global. Reversal couples symmetric digit positions, while carries propagate between adjacent columns, and realizability requires every digit of S to occur both in the input and in the output, with no other digit occurring. This combination naturally leads to a two-sided carry automaton rather than to independent column conditions. For fixed k , the automaton has only finitely many symbolic affine types, which brings Presburger arithmetic and eventual quasipolynomiality into the problem. At the same time, the special carry identity

$$x + (n-1) + 1 = x + n$$

permits neutral extensions of small essential configurations. This produces a finite-core structure whose affine dimensions control the growth of $T(n, k)$ and whose dominant strata determine the leading asymptotics.

Example. In base 10, take

$$S = \{1, 2, 9\}, \quad N = (21999)_{10}.$$

Then

$$\text{Rev}_{10}(N) = (99912)_{10}$$

and

$$(21999)_{10} + (99912)_{10} = (121911)_{10}.$$

Thus both N and $N + \text{Rev}_{10}(N)$ use exactly the digits $\{1, 2, 9\}$, so S is realizable. Column by column, the addition passes through the equations

$$9 + 2 = 1 + 10, \quad 9 + 1 + 1 = 1 + 10, \quad 9 + 9 + 1 = 9 + 10,$$

$$1 + 9 + 1 = 1 + 10, \quad 2 + 9 + 1 = 2 + 10,$$

followed by the final carry 1. This small example already illustrates the interaction between reversal, symmetric digit positions, and carries that the two-sided automaton records.

The first global results are the following.

Theorem 1.2. *For every fixed k , the function $n \mapsto T(n, k)$ is eventually quasipolynomial.*

Theorem 1.3 (Dense-set binomial tail). *For every $n \geq 4$ and*

$$k \geq \left\lceil \frac{n}{2} \right\rceil + 1,$$

every k -element subset of $\{1, \dots, n-1\}$ is realizable. Consequently

$$T(n, k) = \binom{n-1}{k}.$$

Moreover, every such set has a witness N whose base- n expansion has length at most $4k+4$.

Theorem 1.4 (Computer-assisted degree law). *For every fixed $k \geq 3$,*

$$T(n, k) = \Theta(n^{k-3}).$$

Consequently the eventual quasipolynomial of Theorem 1.2 has degree exactly $k-3$.

Theorem 1.5 (Computer-assisted leading coefficient). *For every fixed $k \geq 4$, as $n \rightarrow \infty$ through odd integers,*

$$T(n, k) = \frac{k^2 + k - 4}{4(k-3)!} n^{k-3} + O_k(n^{k-4}),$$

whereas along even integers,

$$T(n, k) = \frac{k(k+1)}{4(k-3)!} n^{k-3} + O_k(n^{k-4}).$$

The support bound of Theorem 5.3 is the key structural step: it converts the classification of primitive accepted mechanisms of fixed rank from an a priori unbounded-support problem into a finite-support problem.

The finite computational dependencies of Theorem 1.5 are isolated explicitly in Sections 9 and 15: the rank-three parts of the exact $k = 4, 5$ classifications identify the support- ≤ 5 macro-families, and Proposition 9.1 excludes new primitive rank-three supports 6 and 7. All subsequent leading-volume and overlap calculations are mathematical deductions from these finite certificates.

The column $k = 3$ is exceptional: it is eventually periodic rather than represented by the neutral-lift transform underlying Theorem 1.5.

The paper is organized as follows. Section 2 develops the two-sided carry automaton, and Section 3 derives eventual quasipolynomiality from its finite symbolic state space. Sections 4–5 introduce neutral layers and primitive cores and establish the structural support bound. Sections 6–9 derive the degree law, the dense-set theorem, exact neutral lifting, and the leading coefficient. Sections 11–13 give the exact low-column classifications, while Sections 14–15 record the finite-kernel consequence and the computational certification framework.

2 The two-sided carry automaton

Write

$$N = (a_{L-1} \cdots a_0)_n.$$

Let c_j be the carry entering column j , with $c_0 = 0$. Since

$$a_j + a_{L-1-j} + c_j \leq 2(n-1) + 1 = 2n-1,$$

every carry lies in $\{0, 1\}$. If b_j is the output digit in column j , then

$$a_j + a_{L-1-j} + c_j = b_j + nc_{j+1}. \quad (1)$$

We process symmetric columns simultaneously. Suppose the current outer carries are $(\lambda, \mu) = (c_j, c_{L-j})$ and the next inner carries are $(\lambda', \mu') = (c_{j+1}, c_{L-j-1})$. If the two input digits are x_p, x_q and the two output digits are x_u, x_v , the corresponding pair layer is

$$x_p + x_q + \lambda = x_u + n\lambda', \quad (2)$$

$$x_p + x_q + \mu' = x_v + n\mu. \quad (3)$$

Thus every local equation has the form

$$x_i + x_j + \varepsilon = x_\ell + \delta n, \quad \varepsilon, \delta \in \{0, 1\}. \quad (4)$$

After cancellation of repeated digit variables, the digit support of a nonzero local row has size at most three.

For a fixed k , order the digits as

$$1 \leq x_1 < \cdots < x_k \leq n-1.$$

An affine row is represented in the coordinates

$$(x_1, \dots, x_k, n, 1).$$

A symbolic state is

$$(\lambda, \mu, I, O, E),$$

where $\lambda, \mu \in \{0, 1\}$, $I, O \subseteq \{1, \dots, k\}$ are the input and output masks, and E is the rational row space generated by the local affine equations used so far.

There are two possible outer boundary states. If the final carry is zero, the initial state is

$$(0, 0, \emptyset, \emptyset, \{0\}).$$

If the final carry is one, ordinary addition produces one additional most significant output digit, and that digit is exactly 1. Since the output digit set must equal S , we must have $1 \in S$; under the ordering above this is equivalent to $x_1 = 1$. Accordingly the carry-one initial state includes the affine row $x_1 - 1 = 0$ and marks digit 1 in the output mask before any paired layer is processed.

If a central digit remains, it satisfies

$$2x_i + \lambda = x_j + n\mu. \quad (5)$$

For even length, the two inner carry fronts must meet. Acceptance requires

$$I = O = \{1, \dots, k\}.$$

Proposition 2.1 (Automaton equivalence). *A k -element set $S \subseteq \{1, \dots, n-1\}$ is realizable if and only if its ordered digits satisfy the equations and inequalities associated with an accepting path of the two-sided automaton.*

Proof. Given a witness N , pair the outermost columns, then the next pair, and so on. Equation (1) gives exactly (2) and (3); an unpaired centre gives (5). The masks are full exactly because the digit sets of input and output both equal S .

Conversely, an accepting path specifies the input digits in symmetric positions together with all adjacent carries. Reading the pair layers from outside inward constructs a finite input word. Equations (2), (3), and, if needed, (5), verify the addition column by column. Full input and output masks give equality of both digit sets with S . $\square$

3 Finite symbolic types and eventual quasipolynomiality

Lemma 3.1 (Finite symbolic state space). *For every fixed k , only finitely many symbolic states can occur.*

Proof. There are finitely many choices for the two carry bits and the two masks. There are also only finitely many possible local affine rows, since the digit indices range over a fixed k -element set and all carry bits are binary. Let $\mathcal{R}_k$ denote this finite row set. Every state row space is generated by a subset of $\mathcal{R}_k$, so only finitely many rational row spaces can occur. $\square$

Lemma 3.2 (Path shortening). *Every accepting symbolic type has an accepting path with no repeated full symbolic state.*

Proof. Suppose an accepting path visits the same full state (λ, μ, I, O, E) twice, and delete the pair-layer segment between those two occurrences. The boundary carries agree, so the surviving pair layers concatenate. The input and output masks also agree at the two occurrences. Since masks are updated only by taking unions, every digit that appears in the deleted segment already appeared before the first occurrence in the same required input or output role. Thus the deleted segment cannot be the unique source of any digit occurrence needed for final acceptance.

Likewise the accumulated row space is E at both occurrences. Since row spaces can only grow by adjoining local rows, every affine consequence contributed inside the deleted segment already lies in E at the first occurrence; deleting the segment therefore loses no affine constraint needed by the continuation. Finally, each deleted step represents a symmetric pair of columns, so the parity of the input length is unchanged and the same centre rule, if any, remains applicable. Hence the shortened path is still accepting. Repeating the operation produces an accepting path with no repeated full state. $\square$

Proof of Theorem 1.2. By Lemmas 3.1 and 3.2, for fixed k only finitely many simple accepting path types need be considered. For each such path, the conditions on $(n, x_1, \dots, x_k)$ are finite systems of integral linear equalities and inequalities, including

$$1 \leq x_1 < \dots < x_k \leq n-1.$$

Taking the finite disjunction over all accepting paths gives a Presburger formula with free parameter n and counted variables $x_1, \dots, x_k$. The strict ordering gives a unique tuple for each k -element digit set. Thus its counting function is exactly $T(n, k)$.

Woods' Presburger counting theorem states that such a counting function is piecewise quasipolynomial. In one free parameter this is, in particular, quasipolynomial for all sufficiently large n [1]. $\square$

4 Neutral layers, essential cores, and rank

The only way a digit can occur in a local equation with net coefficient zero is rigid.

Lemma 4.1 (Unique neutral layer). *Suppose a digit x cancels from a local equation. Then the equation is necessarily*

$$x + (n - 1) + 1 = x + n. \quad (6)$$

Proof. After cancelling x , the equation has the form

$$y + \varepsilon = \delta n, \quad 1 \leq y \leq n - 1, \quad \varepsilon, \delta \in \{0, 1\}.$$

The case $\delta = 0$ is impossible. Thus $\delta = 1$. If $\varepsilon = 0$, then $y = n$, also impossible. Hence $\varepsilon = 1$ and $y = n - 1$, giving (6). $\square$

Thus $n - 1$ is the unique universal *neutral lift anchor*.

Definition 4.2. A digit is *neutral* in an accepted type if it can be deleted while being used only through layers of the form (6). After deleting all neutral digits, the remaining accepted mechanism is its *essential core*. A core is *primitive* if no proper essential submechanism is already accepted.

Let r denote the rank of the full affine row space in the coordinates $(x_1, \dots, x_k, n, 1)$. Let r_{dig} be the rank of the submatrix consisting only of the digit columns.

Lemma 4.3 (Eventual digit rank). *If an affine symbolic type occurs for infinitely many bases n , then*

$$r_{\text{dig}} = r.$$

Proof. Certainly $r_{\text{dig}} \leq r$. If $r_{\text{dig}} < r$, a nonzero rational linear combination of independent affine rows cancels all digit coefficients. It leaves a nonzero relation

$$\alpha n + \beta = 0.$$

This can hold for at most one integer n , contradicting occurrence for infinitely many bases. $\square$

Therefore an eventual rank- r type on s essential digits has fixed- n affine dimension $s - r$.

Lemma 4.4 (Crude compression). *Every accepted rank- r type is a neutral lift of an essential core on at most $3r$ digits.*

Proof. Choose r actual local rows spanning the full row space E . Their digit supports contain at most $3r$ coordinates. By definition, every local row occurring anywhere along the accepting path belongs to E . Hence a digit outside the support union of the chosen basis rows has coefficient zero not only in the basis but in every local row of the path. Every syntactic occurrence of that digit therefore cancels inside its local equation. By Lemma 4.1, such a digit can occur only through the neutral layer (6). Delete all such digits. $\square$

5 Primitive-core compression

We now sharpen Lemma 4.4 for ranks at least three.

Lemma 5.1 (Small-support vectors). *Let $r \geq 3$, and let $\rho_1, \dots, \rho_r$ be actual local rows forming a basis of a primitive core row space E . Let S_i be the digit support of ρ_i . If*

$$\left| \bigcup_{i=1}^r S_i \right| \geq 3r - 1,$$

then every vector of E having digit support at most three is proportional to a single basis row.

Proof. Each $|S_i| \leq 3$. If the union has size at least $3r - 1$, the total deficit from the maximum $3r$ is at most one. Hence only three support patterns are possible:

- (i) all S_i have size three and are pairwise disjoint;
- (ii) all have size three, exactly two meet in one coordinate, and all other intersections are empty;
- (iii) one S_i has size two and all remaining supports have size three, with all supports disjoint.

A nontrivial combination of two disjoint basis rows has digit support at least five. In case (ii), cancellation can occur only at the unique shared coordinate, leaving at least four private coordinates. Additional basis rows cannot cancel coordinates that occur uniquely in their supports. Thus a combination involving at least two basis rows cannot have digit support at most three. $\square$

Lemma 5.2 (Proportional rows preserve digit roles). *Two proportional nonzero local rows of digit support two or three have the same input/output roles.*

Proof. A support-three digit row has coefficient pattern

$$e_a + e_b - e_c,$$

with c the output coordinate. Two such local rows can be proportional only with scalar one, so they have the same output and the same unordered input pair. A support-two local row has pattern $2e_a - e_c$, and the same argument applies. $\square$

Theorem 5.3 (Primitive-core support bound). *Let $r \geq 3$. Every primitive accepted rank- r core has at most*

$$\boxed{3r - 2}$$

essential digits.

Proof. Assume a primitive accepted core has at least $3r - 1$ essential digits. Every essential digit has nonzero coefficient in some local relation, so the union of the supports of a basis of actual local rows contains all essential digits. Lemma 5.1 applies. Under the present large-support hypothesis, every basis row has digit support two or three, so no support-one vector can be proportional to a basis row.

Since $r \geq 3$, in each of the three support patterns in the proof of Lemma 5.1 there exists a size-three basis support $\{a, b, c\}$ disjoint from every other basis support. Write the corresponding local equation as

$$x_a + x_b + \varepsilon = x_c + \delta n.$$

The digit x_c is an output digit. By Lemmas 5.1 and 5.2, every nonneutral local row containing x_c lies on the same basis line and gives x_c the same output role. Hence x_c can never occur as a nonneutral input.

Could it occur neutrally? Lemma 4.1 shows that a neutral occurrence would force the support-one affine row

$$x_j + 1 - n = 0$$

for the anchor digit $x_j = n - 1$. But Lemma 5.1 says that every row of E with digit support at most three is proportional to one of the basis rows, all of which have digit support two or three under the present hypothesis. Hence E contains no support-one local row. A neutral rescue is impossible.

Thus x_c never occurs as an input digit, contradicting acceptance of the essential core. Therefore $s \leq 3r - 2$. $\square$

Corollary 5.4. *Every primitive rank-three core has at most seven essential digits.*

6 Minimum eventual rank and the degree law

Ranks one and two are small enough to be eliminated by a finite exact certification.

Proposition 6.1 (Computer-assisted low-rank exclusion). *There is no accepted essential core of rank at most two that occurs for all sufficiently large bases.*

Certification. By Lemma 4.4, any rank- $r \leq 2$ accepted type reduces to an essential core on at most six digits. The exact classifications for $k \leq 5$ contain no accepted system of rank below three. A standalone rank-filtered symbolic search for $k = 6$ retains only states of rank at most two; since rank cannot decrease when new local equations are appended, this is an exhaustive search for all such accepted systems. The recorded exact Maple run terminates with no accepted system. All row reduction and feasibility operations are rational and exact. See Appendix B for the certification dependency. $\square$

Proof of Theorem 1.4. For the lower bound, the three-digit core

$$\{1, 2, n - 1\}$$

has an accepting witness containing a carry- $(1, 1)$ region. At such a region, any additional digit x may be inserted through

$$x + (n - 1) + 1 = x + n,$$

without adding a new independent relation. Thus every k -set containing $\{1, 2, n - 1\}$ is realizable for sufficiently large n . There are

$$\binom{n-4}{k-3} = \frac{n^{k-3}}{(k-3)!} + O_k(n^{k-4})$$

such sets.

For the upper bound, Section 3 gives only finitely many eventual simple symbolic path types. By Proposition 6.1, every eventual accepted type has rank at least three. By Lemma 4.3, its fixed- n affine dimension is at most $k - 3$. A fixed rational affine set of dimension d contains $O(n^d)$ integer points inside the box $1 \leq x_1 < \dots < x_k \leq n - 1$. Summing over finitely many types gives

$$T(n, k) = O_k(n^{k-3}).$$

The two estimates prove $\Theta(n^{k-3})$. Theorem 1.2 then implies that the eventual quasipolynomial has degree exactly $k - 3$. $\square$

7 Dense digit sets and the binomial tail

The proof is constructive. Put $m = n - 1$ and represent the cyclic group $\mathbb{Z}/m\mathbb{Z}$ by $\{1, \dots, m\}$, with m representing the zero residue. For an integer q , let $\langle q \rangle_m$ denote this representative.

Lemma 7.1 (Cyclic overlap). *Let $S \subseteq \{1, \dots, m\}$ with $|S| > m/2$.*

(i) *For every $z \in S$ there exist $x, y \in S$ such that*

$$\langle x + y \rangle_m = z.$$

(ii) *For every $x \in S$ there exists $y \in S$ such that*

$$\langle x + y \rangle_m \in S.$$

Proof. For (i), in $\mathbb{Z}/m\mathbb{Z}$ the sets S and $z - S$ both have cardinality $|S| > m/2$, so they intersect. If $x \in S \cap (z - S)$, then $y = z - x$ modulo m belongs to S , and $x + y \equiv z \pmod{m}$.

For (ii), the sets S and $S - x$ again both have cardinality greater than $m/2$, so choose $y \in S \cap (S - x)$. Then $x + y$ modulo m belongs to S . $\square$

A pair $(x, y) \in S^2$ will be called *low* if $x + y \leq m$, and *high* if $x + y > m$. A low pair with $\langle x + y \rangle_m = z$ gives a carry-(0, 0) loop: both symmetric output digits are $z = x + y$ and both next carries remain zero. A high pair gives a carry-(1, 1) loop: since $n = m + 1$,

$$x + y + 1 = (x + y - m) + n,$$

so both symmetric output digits are $z = x + y - m = \langle x + y \rangle_m$ and both carries remain one.

To pass from the low-loop region to the high-loop region we need two bridge layers. Their existence follows from the next elementary density lemma.

Lemma 7.2 (Low and high bridges). *Let $m \geq 3$ and let $S \subseteq \{1, \dots, m\}$ satisfy*

$$|S| \geq \left\lfloor \frac{m}{2} \right\rfloor + 2.$$

Then there exist $a, b, x_0, y_0, x_1, y_1 \in S$, with $a, a + 1, b, b + 1 \in S$, such that

$$x_0 + y_0 = a \quad \text{and} \quad x_1 + y_1 = m + 1 + b.$$

Proof. Since $|S| > \lceil m/2 \rceil$, the set S contains two consecutive integers.

First suppose that no low bridge exists. Let a be the largest integer with $a, a+1 \in S$. Because a has no representation $x+y=a$ with $x, y \in S$, the elements of $S \cap \{1, \dots, a-1\}$ contain at most one member of each complementary pair $\{u, a-u\}$; if a is even, the fixed point $a/2$ cannot be selected. Hence

$$|S \cap \{1, \dots, a-1\}| \leq \left\lfloor \frac{a-1}{2} \right\rfloor.$$

By maximality of a , one has $a+2 \notin S$, and there are no consecutive selected integers above $a+1$. Therefore

$$|S \cap \{a+2, \dots, m\}| \leq \left\lceil \frac{m-a-2}{2} \right\rceil,$$

where the right side is interpreted as zero when the interval is empty. Moreover,

$$\left\lfloor \frac{a-1}{2} \right\rfloor + \left\lceil \frac{m-a-2}{2} \right\rceil \leq \left\lceil \frac{m-3}{2} \right\rceil = \left\lfloor \frac{m}{2} \right\rfloor - 1.$$

Therefore

$$|S| \leq \left\lfloor \frac{m}{2} \right\rfloor + 1,$$

contrary to the hypothesis. Hence a low bridge exists.

Now suppose that no high bridge exists. Let b be the smallest integer with $b, b+1 \in S$. By minimality, $b-1 \notin S$ when $b > 1$, and below b there are no consecutive selected integers. Hence

$$|S \cap \{1, \dots, b-1\}| \leq \left\lceil \frac{b-2}{2} \right\rceil.$$

Consider the interval $I = \{b+1, \dots, m\}$. The involution

$$t \mapsto m+1+b-t$$

maps I to itself. Since no two selected elements may sum to $m+1+b$, at most one member of each two-cycle belongs to S , while a fixed point, if present, cannot belong to S . Thus

$$|S \cap I| \leq \left\lfloor \frac{m-b}{2} \right\rfloor.$$

Moreover,

$$\left\lceil \frac{b-2}{2} \right\rceil + \left\lfloor \frac{m-b}{2} \right\rfloor \leq \left\lceil \frac{m-2}{2} \right\rceil = \left\lceil \frac{m}{2} \right\rceil - 1.$$

Including b itself therefore gives

$$|S| \leq \left\lceil \frac{m}{2} \right\rceil.$$

But $\lfloor m/2 \rfloor + 2 > \lceil m/2 \rceil$, a contradiction. Hence a high bridge exists as well. $\square$

Proof of Theorem 1.3. Let $m = n - 1$ and let $S \subseteq \{1, \dots, m\}$ have cardinality

$$k \geq \left\lfloor \frac{n}{2} \right\rfloor + 1 = \left\lfloor \frac{m}{2} \right\rfloor + 2.$$

In particular $k > m/2$.

For each output digit $z \in S$, Lemma 7.1(i) supplies a pair $(x_z, y_z) \in S^2$ with $\langle x_z + y_z \rangle_m = z$. Keep one such loop for every z . For each input digit $x \in S$, Lemma 7.1(ii) supplies

a pair $(x, y_x) \in S^2$ whose cyclic output $\langle x + y_x \rangle_m$ belongs to S . Keep one such loop for every x . Classify all these loops as low or high according as their pair sum is at most m or greater than m .

By Lemma 7.2, choose a low bridge

$$x_0 + y_0 = a, \quad a, a + 1 \in S,$$

and a high bridge

$$x_1 + y_1 = n + b, \quad b, b + 1 \in S.$$

Now concatenate, from the outside toward the center,

all low loops, the low bridge, the high bridge, all high loops.

The carry states evolve as

$$(0, 0) \longrightarrow (0, 0) \longrightarrow (0, 1) \longrightarrow (1, 1) \longrightarrow (1, 1).$$

Indeed, a low loop of sum $q \leq m$ outputs q on both sides and stays at $(0, 0)$. For the low bridge,

$$a + 0 = a + n \cdot 0, \quad a + 1 = (a + 1) + n \cdot 0,$$

so the next inner carry state is $(0, 1)$. For the high bridge,

$$n + b + 0 = b + n, \quad n + b + 1 = (b + 1) + n,$$

so the next state is $(1, 1)$. Finally every high loop of sum $q > m$ satisfies

$$q + 1 = (q - m) + n$$

on both sides and therefore stays at $(1, 1)$. The two inner carries are equal at the end, so this is an even-length accepting path.

Every digit of S occurs as an output because of the first collection of loops, and every digit of S occurs as an input because of the second collection. All loop and bridge inputs belong to S , and all their outputs belong to S . Hence both the input and output masks are exactly S . By Proposition 2.1, the path gives a witness N , so S is realizable.

There are at most k output-covering loops, k input-covering loops, and two bridge layers. Thus there are at most $2k + 2$ symmetric pair layers and the resulting witness may be chosen with at most $4k + 4$ base- n digits.

Since every k -subset is realizable and there are exactly $\binom{n-1}{k}$ such subsets,

$$T(n, k) = \binom{n-1}{k}.$$

□

8 Exact neutral lifting

Definition 8.1. A primitive accepted core is *neutral-liftable* if one of its accepting witnesses contains a carry- $(1, 1)$ region into which arbitrarily many layers of the form (6) may be inserted.

Such a core necessarily contains the anchor $n - 1$.

Proposition 8.2 (Exact lift multiplicity). *Let C be a fixed neutral-liftable core instance containing s distinct digits. For $m \geq 0$, the number of $(s + m)$ -element digit sets obtained by adjoining m neutral digits to C is exactly*

$$\boxed{\binom{n-1-s}{m}}.$$

Proof. There are $n - 1 - s$ nonzero digit values outside C . Choose any m of them. Insert, for each chosen digit x , one neutral carry-(1, 1) layer

$$x + (n - 1) + 1 = x + n.$$

Each new digit then occurs in both the input and output masks, while the core row space is unchanged. Conversely, the lifted digit set uniquely determines the chosen m -subset of the complement. $\square$

The exact binomial factor gives

$$\binom{n-1-s}{m} = \frac{n^m}{m!} - \frac{ms + \frac{1}{2}m(m+1)}{m!} n^{m-1} + O_{s,m}(n^{m-2}). \quad (7)$$

In particular, neutral lifting multiplies the leading volume by $1/m!$.

Remark 8.3. The exact factor in Proposition 8.2 counts lifts of a fixed core instance. When several core families are united, overlaps must be treated separately. For the leading rank-three layer, this is done in Lemma 9.2 below. Lower asymptotic coefficients require more care and are not claimed here to follow from a naive sum of lift expansions.

9 Rank-three cores and the leading coefficient

Corollary 5.4 reduces primitive rank-three cores to supports at most seven. Two exact finite certifications finish the support-exclusion part of the dominant classification. The support- ≤ 5 geometric macro list is obtained from the exact rank-three parts of the $k = 4$ and $k = 5$ classifications; once that list is fixed, the only additional computational input needed to prove that no further dominant support occurs is Proposition 9.1 below. The leading masses of the displayed macros are then counted elementarily.

Proposition 9.1 (Computer-assisted rank-three support classification). *No new primitive accepted rank-three core has support six or seven. Consequently every primitive rank-three core has support at most five.*

Certification. The exact $k = 6$ rank-three audit finds 25 coordinate-labelled minimal rank-three row spaces, with essential-support distribution

$$6 \text{ on support } 3, \quad 15 \text{ on support } 4, \quad 4 \text{ on support } 5, \quad 0 \text{ on support } 6.$$

A separate primitive $k = 7$ audit freezes every rank-three row space whose digit support is all seven digits and completes it by a finite carry/mask search using only implied equations. The recorded run finds zero accepted support-seven systems. Together with Corollary 5.4, this excludes every new primitive rank-three core of support at least six. $\square$

The exact rank-three classifications for the columns $k = 4$ and $k = 5$ identify the neutral-liftable families below. They are most naturally described geometrically. For support three the liftable cores are

$$\{1, 2, n-1\}, \quad \{1, n-2, n-1\},$$

and, when n is even,

$$\left\{ \frac{n}{2} - 1, n-2, n-1 \right\}.$$

Thus the support-three leading masses are

$$p_3^{\text{odd}} = 2, \quad p_3^{\text{even}} = 3. \quad (8)$$

There are three one-parameter support-four macro-families:

$$\{1, q, q+1, n-1\}, \quad (9)$$

$$\{a, 2a, 2a+1, n-1\}, \quad (10)$$

$$\{a, b, n-2, n-1\}, \quad a+b = n-2, \quad a < b. \quad (11)$$

The first has $n + O(1)$ instances and the other two have $n/2 + O(1)$ instances. Their aggregate support-four mass is therefore

$$p_4 = 2. \quad (12)$$

The support-five macro-family is

$$G_2(a, b) = \{a, b, a+b, a+b+1, n-1\}, \quad 1 \leq a < b, \quad a+b \leq n-3. \quad (13)$$

Its exact number of instances is

$$\left\lfloor \frac{(n-4)^2}{4} \right\rfloor, \quad (14)$$

so its normalized leading mass is

$$p_5 = \frac{1}{4}. \quad (15)$$

Indeed, summing over $s = a + b$ gives

$$\sum_{s=3}^{n-3} \left\lfloor \frac{s-1}{2} \right\rfloor = \left\lfloor \frac{(n-4)^2}{4} \right\rfloor.$$

Lemma 9.2 (Top-dimensional overlap). *After geometrically identical rank-three strata have been merged, the intersection of two distinct neutral-lifted dominant strata contributes $O_k(n^{k-4})$ k -element digit sets.*

Proof. Work in the ordered digit coordinates $x_1 < \dots < x_k$. Every dominant rank-three lifted stratum lies in a rational affine space of codimension three, by Lemma 4.3. Let A_1, A_2 be the affine hulls of two strata. If their rank-three equation spaces were equal, the strata would have the same affine hull and would have been merged before volume addition. Otherwise the span of the two equation spaces has rank at least four. Hence, if nonempty, $A_1 \cap A_2$ has codimension at least four. It contains $O_k(n^{k-4})$ integer points in the relevant box. Lower-dimensional boundaries satisfy the same estimate. $\square$

Proof of Theorem 1.5. By Proposition 9.1, every dominant primitive accepted core has support three, four, or five. Proposition 8.2 shows that a support- s family lifted to k digits contributes a factor

$$\binom{n-1-s}{k-s} = \frac{n^{k-s}}{(k-s)!} + O_k(n^{k-s-1}).$$

Combining this with the support- s core volumes gives, for odd n ,

$$c_k^{\text{odd}} = \frac{2}{(k-3)!} + \frac{2}{(k-4)!} + \frac{1}{4(k-5)!},$$

where terms with negative factorial arguments are omitted. For $k \geq 5$ this simplifies to

$$c_k^{\text{odd}} = \frac{k^2 + k - 4}{4(k-3)!},$$

and the same closed expression also gives $c_4^{\text{odd}} = 4$.

For even n , the additional support-three core contributes $1/(k-3)!$, so

$$c_k^{\text{even}} = c_k^{\text{odd}} + \frac{1}{(k-3)!} = \frac{k(k+1)}{4(k-3)!}.$$

Lemma 9.2 shows that dominant-stratum intersections affect only $O_k(n^{k-4})$. Every rank at least four also contributes $O_k(n^{k-4})$ by Lemma 4.3. This proves the theorem. $\square$

Corollary 9.3 (Normalized leading coefficients). *For $k \geq 4$,*

$$(k-3)! c_k^{\text{odd}} = \frac{k^2 + k - 4}{4}, \quad (k-3)! c_k^{\text{even}} = \frac{k(k+1)}{4}.$$

In particular,

$$(k-3)!(c_k^{\text{even}} - c_k^{\text{odd}}) = 1.$$

The correct exponential lift transforms begin at $k = 4$:

$$\sum_{m \geq 1} c_{m+3}^{\text{odd}} z^m = \left(2 + 2z + \frac{1}{4}z^2\right) e^z - 2, \quad (16)$$

$$\sum_{m \geq 1} c_{m+3}^{\text{even}} z^m = \left(3 + 2z + \frac{1}{4}z^2\right) e^z - 3. \quad (17)$$

The omitted constant terms reflect the nonliftable zero-dimensional families present in the three-digit column.

10 Counting conventions for symbolic and geometric objects

The computer-assisted classification naturally produces several different object counts.

Definition 10.1. For fixed k and rank r , let $\mathcal{R}_{k,r}$ be the set of globally row-space-minimal accepted affine systems in the ordered coordinates $x_1 < \cdots < x_k$.

These systems retain coordinate labels. The *essential digit support* of $E \in \mathcal{R}_{k,r}$ is the set of digit columns on which some row of E has nonzero coefficient. A *geometric macro-family* is the union of all ordered coordinate chambers and symbolic row spaces that describe the same parametric family of unordered digit sets.

Thus there are three distinct levels:

$$\begin{aligned} \text{symbolic paths} &\longrightarrow \text{minimal coordinate-labelled row spaces} \\ &\longrightarrow \text{geometric macro-families.} \end{aligned}$$

Their cardinalities need not agree.

For example, the $k = 5$ rank-three search gives 14 minimal coordinate-labelled row spaces, whereas their dominant realizable-set union is described by seven geometric macro-families. The $k = 6$ rank-three audit gives 25 minimal coordinate-labelled row spaces, with support multiplicities 6, 15, 4, 0. These numbers are therefore not contradictory, and the phrase “25 primitive rank-three cores” should be avoided without qualification.

11 The first three columns

Proposition 11.1. *For every $n \geq 2$,*

$$T(n, 1) = 0.$$

Proof. If the unique digit is a , the units column has incoming carry zero and would have to satisfy $2a \equiv a \pmod{n}$. Since $1 \leq a < n$, this is impossible. $\square$

Proposition 11.2. *For $n \geq 3$,*

$$T(n, 2) = \begin{cases} 1, & n = 3 \text{ or } 5, \\ 0, & \text{otherwise.} \end{cases}$$

The realizable cases are

$$(22)_3 + (22)_3 = (121)_3$$

and

$$(313)_5 + (313)_5 = (1131)_5.$$

Proof. Let $S = \{a, b\}$ with $1 \leq a < b < n$. A mixed units pair $a + b$ cannot output either a or b : if no carry is produced, equality with a or b forces the other input digit to be zero; if a carry is produced, equality forces an input digit to be n . Thus the outer pair is aa or bb .

For aa , legality forces $b = 2a$ and outgoing carry zero. A legal interior use of b would have to employ bb ; the only carry-producing exceptional relation is $n = 3a$, and direct examination of the resulting two carry states shows that no mirrored completion can return to carry zero. Thus this branch is impossible.

For bb , legality forces $a = 2b - n$ and outgoing carry one. The final carry is then one, so the leading output digit 1 belongs to S . Hence $a = 1$ and $n = 2b - 1$. If $b \geq 4$, the possible pairs $11, 1b, bb$ with incoming carry one output respectively $3, b + 2, 2$, none in $\{1, b\}$. Thus $b = 2$ or 3 , yielding $n = 3$ and 5 . $\square$

Theorem 11.3 (Three-digit classification). *For every $n \geq 14$, the realizable three-element digit sets are exactly*

$$A_n = \{1, 2, n - 1\}, \quad B_n = \{1, n - 4, n - 2\}, \quad C_n = \{1, n - 2, n - 1\},$$

together with

$$D_a = \{a, 2a, 2a + 1\}$$

when either $n = 2a + 2$ or $n = 3a + 1$. Consequently

$$\boxed{T(n, 3) = 3 + [2 \mid n] + [n \equiv 1 \pmod{3}]} \quad (n \geq 14).$$

Proof. We give the structural reduction; the finite transition table used at the last step is reproduced in Appendix A.

Let $S = \{a, b, c\}$ with $a < b < c$. First suppose the final carry is zero. Any input occurrence of c must occur in a high column, since otherwise its sum with another positive digit would output a digit larger than c . Strip matching outer carry- $(0, 0)$ layers until reaching the first layer whose low-side occurrence has incoming carry zero and whose mirrored high-side occurrence has incoming carry one. If the pair sum is q , the two outputs are q and $q + 1$, both in S . Since $q \geq 2a > a$, they must be b and c . The output b can only come from the pair aa , so

$$b = 2a, \quad c = 2a + 1.$$

Requiring c itself to occur as an input digit leaves only

$$n \in \{2a + 2, 2a + 3, 3a + 1, 3a + 2, 3a + 3\}.$$

The transition table excludes $2a + 3$, $3a + 2$, and $3a + 3$; the two survivors give the family D_a and have explicit witnesses.

Now suppose the final carry is one. Then $1 \in S$, so write $S = \{1, b, c\}$. The outer units pair is high and cannot involve 1. Thus it is bb , bc , or cc , giving the outer relations

$$n = 2b - 1, \quad n = b + c - 1, \quad n = 2c - 1, \quad n = 2c - b.$$

If the mirrored outer occurrence has incoming carry one, the two outer outputs are consecutive; the surviving possibilities are A_n and C_n . Otherwise an interior $1 \rightarrow 0$ transition is required. For $n \geq 14$ the only possible drop types are

$$11 \ (b = 3), \quad 1b \ (c = b + 2), \quad bb \ (c = 2b + 1).$$

Combining these with the four outer relations gives a finite table. All cases fail an input/output role condition except

$$n = 2c - b, \quad c = b + 2,$$

which yields $b = n - 4$, $c = n - 2$ and hence B_n . Explicit digit strings verify all listed families. $\square$

For reference, the small values for $n = 4, \dots, 13$ are

$$1, 3, 4, 5, 4, 5, 5, 4, 4, 5.$$

12 The fourth column

Theorem 12.1 (Computer-assisted exact fourth column). *For every $n \geq 30$,*

$$\begin{aligned} T(n, 4) = & 4n - 6 + (n - 8)[2 \mid n] + [n \equiv 1 \pmod{3}] \\ & + 2[n \bmod 4 \in \{1, 2\}] + c_5(n) + c_7(n), \end{aligned} \tag{18}$$

where

$$c_5(n) = \begin{cases} 5, & n \equiv 1, 3 \pmod{5}, \\ 4, & n \equiv 2 \pmod{5}, \\ 0, & n \equiv 0, 4 \pmod{5}, \end{cases}$$

and

$$c_7(n) = \begin{cases} 1, & n \equiv 1, 2, 3, 4, 5 \pmod{7}, \\ 0, & n \equiv 0, 6 \pmod{7}. \end{cases}$$

The eventual quasipolynomial has exact period 420. The threshold 30 is sharp: $T(29, 4) = 114$, whereas (18) gives 113.

Certification and count. The exact symbolic automaton terminates with precisely 60 globally minimal accepted affine row spaces: seven of rank three and 53 of rank four. The seven rank-three solution forms are

$$\begin{array}{ll} (1, 2, c, n-1), & (1, b, n-2, n-1), \\ (n-b-2, b, n-2, n-1), & (n/2-1, b, n-2, n-1), \\ (1, c-1, c, n-1), & ((c-1)/2, c-1, c, n-1), \\ (a, n/2-1, n-2, n-1), & \end{array}$$

where the fourth and seventh occur only for even n . After strict ordering and removal of their boundary overlaps, their union has size

$$U_3(n) = 4n - 20 + (n-6)[2 \mid n].$$

Each rank-four system gives one ordered affine four-set in the congruence classes where its coordinates are integral and ordered. Counting the 53 exact families gives

$$K(n) = 12 + 2[n \text{ odd}] + [n \equiv 1 \pmod{3}] + 2[n \bmod 4 \in \{1, 2\}] + c_5(n) + c_7(n).$$

For $n \geq 30$, these rank-four families are pairwise distinct and disjoint from the rank-three union, so $T(n, 4) = U_3(n) + K(n)$, which simplifies to (18).

The residue pattern has no proper period dividing $\text{lcm}(2, 3, 4, 5, 7) = 420$. An independent exact direct-subset audit verifies the classified union against all four-element subsets through the recorded test range. All feasibility tests use exact rational Fourier–Motzkin elimination. $\square$

The coefficient of n in Theorem 12.1 is 4 for odd n and 5 for even n , in agreement with Theorem 1.5.

13 The fifth column

The fifth column exhibits the complete rank filtration 3, 4, 5.

Theorem 13.1 (Computer-assisted rank filtration for $k = 5$). *For $n \geq 50$, the symbolic classification contains*

$$14$$

minimal rank-three systems,

$$14$$

genuinely new rank-four systems not contained in a rank-three family, and

$$1112$$

genuinely new rank-five systems. Their fixed- n fiber dimensions are respectively 2, 1, 0.

Certification. The rank-three extractor performs an exact rank-at-most-three search, reduces accepted systems by row-space containment, and obtains 14 minimal rank-three row spaces. Their digit coefficient matrices all have rank three. The rank-four search retains total rank at most four and discards states whose row spaces already contain one of the accepted rank-three systems; it returns 14 new minimal rank-four systems, all of digit rank four. The rank-five search repeats the same procedure after pruning both lower ranks and returns 1112 new minimal systems, all of digit rank five. Since rank cannot decrease along a path, each rank-filtered search is exhaustive within its rank bound. $\square$

The fourteen rank-three row spaces are geometrically represented by seven macros:

M_1 : sets containing $\{1, 2, n - 1\}$;

M_2 : sets containing $\{1, n - 2, n - 1\}$;

M_3 : for even n , sets containing $\{n/2 - 1, n - 2, n - 1\}$;

M_4 : $\{1, q, q + 1, n - 1, x\}$;

M_5 : $\{a, 2a, 2a + 1, n - 1, x\}$;

M_6 : $\{a, b, n - 2, n - 1, x\}$ with $a + b = n - 2$, $a < b$;

M_7 : $\{a, b, a + b, a + b + 1, n - 1\}$ with $a < b$.

This is the geometric origin of the masses used in Theorem 1.5.

Theorem 13.2 (Computer-assisted exact fifth column). *For every $n \geq 50$,*

$$\boxed{T(n, 5) = Q_5(n) + B_5(n)},$$

where

$$Q_5(n) = \begin{cases} \frac{15}{4}n^2 - 43n + 120 + [4 \mid n] + \frac{5n - 32}{3}[n \equiv 1 \pmod{3}], & n \text{ even}, \\ \frac{13n^2 - 144n + 387}{4} + \frac{5n - 32}{3}[n \equiv 1 \pmod{3}], & n \text{ odd}. \end{cases}$$

The remainder B_5 is periodic with minimal period

$$27720 = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11,$$

and satisfies

$$154 \leq B_5(n) \leq 302.$$

Its minimum occurs exactly for

$$n \equiv 0, 25200 \pmod{27720},$$

and its maximum exactly for

$$n \equiv 4461, 9501, 22101, 27141 \pmod{27720}.$$

Its mean over one full minimal period is

$$\frac{6708851}{27720}.$$

Certification and decomposition. Let $R_3(n)$ be the union of the seven rank-three macros, and let $R_4(n)$ be the union of the fourteen explicit rank-four families. An independent contribution audit constructs these two unions directly, without using the stored minimal row-space lists, and obtains

$$|R_4(n) \setminus R_3(n)| = 2n - 14 + [n \bmod 4 \in \{0, 3\}] + \frac{5(n-7)}{3} [n \equiv 1 \pmod{3}].$$

Adding the exact rank-three macro count gives the displayed function $Q_5(n)$.

For the rank-five part, every new system is solved affinely. The exact postprocessor removes duplicate affine families, eventually invalid order chambers, permanent lower-rank coverage, all isolated intersections between distinct rank-five families, and all isolated intersections with lower-rank families. Beyond the certified threshold, the surviving families are distinct and do not meet lower-rank strata. Hence their contribution is exactly the number whose affine coordinates are integral.

The 1112 symbolic rank-five systems contain 1107 arithmetically active affine families; five are sterile. The compressor records 64 congruence signatures in total: 63 nonempty single-residue signatures and one empty signature of multiplicity five. Exact reconstruction over the common denominator period and divisor-period reduction give minimal period 27720. Direct inspection of the exact residue table gives the stated extrema and mean. $\square$

The quadratic coefficients $13/4$ for odd bases and $15/4$ for even bases agree with Theorem 1.5 at $k = 5$.

14 A finite-kernel principle

The support theorem has a useful consequence beyond rank three, although we do not attempt here to compute lower asymptotic coefficients in general.

Corollary 14.1 (Finite primitive kernels at fixed rank). *Fix $r \geq 3$. Every primitive accepted rank- r core has support at most $3r - 2$. Hence, for columns $k > 3r - 2$, every eventual rank- r accepted type is obtained by neutral lifting from a finite collection of anchored primitive accepted core types of bounded support.*

Proof. The support bound is Theorem 5.3. For fixed support there are only finitely many simple symbolic core types by Lemmas 3.1 and 3.2. If $k > 3r - 2$, a rank- r accepted type cannot have all k digits essential, so at least one digit is neutral; Lemma 4.1 identifies the unique neutral mechanism and its anchor. $\square$

Remark 14.2. Corollary 14.1 is deliberately weaker than a full “all coefficients” transform. At lower asymptotic orders, intersections of strata and boundary terms must be organized before one may add individual lift expansions. The leading rank-three layer is exceptional in its simplicity because distinct top-dimensional affine hulls meet in codimension at least four.

15 Computational certification and reproducibility

The exact symbolic searches use the same finite-state semantics as Proposition 2.1. A state stores two carries, two digit masks, and the exact rational row space of all equations used so far. Feasibility is tested against the strict digit ordering using exact rational elimination; no floating-point feasibility decision is used.

15.1 Safe row-space containment reduction

Throughout this section, a *row-space-minimal accepted family* means an accepted row space that is inclusion-minimal among all accepted row spaces for the fixed column k and eventual range under consideration.

For rational row spaces E_1, E_2 in the affine coordinates $(x_1, \dots, x_k, n, 1)$, the implementation tests

$$E_1 \subseteq E_2 \iff \text{rref}(E_2 \cup E_1) = \text{rref}(E_2). \quad (19)$$

This gives the following formal pruning rule.

Proposition 15.1 (Containment pruning). *Fix the same discrete state (λ, μ, I, O) . If two live symbolic states have row spaces $E_1 \subseteq E_2$, then the state carrying E_2 may be deleted without losing any row-space-minimal accepted family.*

Proof. Let a continuation of the stronger state be given by symbolic layers $L_1, \dots, L_m$, and let F_j denote the set of local affine rows contributed by the first j layers. The choice of a symbolic layer depends only on the current discrete carry state and on its input/output labels. Since the two initial states have the same carries and masks, we may attempt the same layer sequence from the state carrying E_1 .

We first record the feasibility monotonicity that makes this legitimate. The strict ordering inequalities

$$1 \leq x_1 < \dots < x_k \leq n - 1$$

and the lower bound on n are identical for the two states. Thus the monotonicity used here comes solely from deleting affine equations while leaving every inequality unchanged. For every prefix j ,

$$\text{span}(E_1, F_j) \subseteq \text{span}(E_2, F_j).$$

Hence every tuple satisfying the stronger prefix system also satisfies the weaker prefix system. Consequently,

$$\text{Feas}(\text{span}(E_2, F_j)) \implies \text{Feas}(\text{span}(E_1, F_j))$$

for every j . Thus no exact consistency or order-feasibility test that allows the stronger continuation can reject the corresponding weaker prefix.

Suppose now that the continuation from E_2 ends in a row-space-minimal accepted row space

$$M_2 = \text{span}(E_2, F_m).$$

Following the same symbolic layers from E_1 reaches the same final carries and the same final input/output masks, so it is accepted as well; its row space is

$$M_1 = \text{span}(E_1, F_m) \subseteq M_2.$$

By definition, M_2 is inclusion-minimal among accepted row spaces. Since M_1 is accepted, this inclusion forces $M_1 = M_2$. Therefore every row-space-minimal accepted family obtainable from the discarded stronger state is also obtainable from the retained weaker state. Deleting the state carrying E_2 is safe. $\square$

Rank-filtered searches are exhaustive because appending equations cannot decrease rank. The support-six and support-seven rank-three audits use the same monotonicity: once rank three is reached, a candidate row space is frozen and only discrete carry/mask completions using rows already implied by that space need be searched.

For clarity, the main logical dependencies are:

Statement	Finite certificate used
Minimum eventual rank 3	exact exclusion of rank ≤ 2 through core support 6
Complete dominant rank-three classification	no new primitive support-6 or support-7 rank-three core
Exact $T(n, 4)$	$60 = 7 + 53$ minimal affine systems and exact family count
Exact $T(n, 5)$	rank counts 14, 14, 1112 and exact residue reconstruction

The accompanying computational supplement contains the complete executable source files for certificates C1–C5, a machine-readable file of expected decisive outputs, a reproduction README, fresh archival transcripts from August 11, 2026, historical recorded outcomes, and SHA-256 hashes for every distributed ancillary file. The convenience source bundle contains the complete source bodies; document-preview systems may display only excerpts of that large file, but those excerpts are not the archival source package. The ancillary archive and the files it contains are authoritative. The C5 raw transcript is preserved together with an explicit postvalidation note because the legacy final driver used for that rerun contained two stale post-run assertions; the underlying exact C5 certificate stages completed with the expected outputs, and the corrected driver and stricter launcher are included in the release.

The structural support theorem, neutral-lift theorem, dimension argument, and Presburger reduction do not depend on numerical experimentation.

16 Questions and further directions

The present paper settles the fixed-column degree and leading coefficient, but several natural questions remain.

Question 16.1. Can the support bound $3r - 2$ be sharpened for accepted primitive cores? Computational evidence in small ranks suggests substantially more rigidity, but no stronger universal bound is used here.

Question 16.2. Determine the complete next coefficient of $T(n, k)$ for arbitrary fixed k . More generally, can one organize intersections of primitive-core strata into a finite lift calculus that gives each fixed asymptotic depth uniformly in k ?

Question 16.3. Classify primitive rank-four cores. This is the first new layer required for a uniform description of the coefficient of n^{k-4} .

Question 16.4. What is the typical behavior when k grows with n rather than remaining fixed? The five-digit family $G_2(a, b)$ is a natural source of mesoscopic phenomena because it contains $\Theta(n^2)$ anchored cores.

Question 16.5. Determine efficient formulas or recurrences for the full two-parameter triangle $T(n, k)$ and its row statistics.

AI-assisted research disclosure

The mathematical exploration and preparation of this manuscript were carried out with substantial assistance from OpenAI’s ChatGPT. The system was used interactively to

explore conjectures, develop and revise proofs, design exact symbolic computations, identify possible gaps, and draft and edit the manuscript. The author selected the research problem and final statements, executed the stated Maple audits, reviewed their outputs, and remains responsible for all mathematical claims.

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A Details for the three-digit classification

We record the finite transition reductions used in Theorem 11.3 so that the theorem does not depend on an unspecified computer search.

For final carry zero, the first mixed-carry outer layer forces

$$S = \{a, 2a, 2a + 1\}.$$

To use $c = 2a + 1$ as an input digit, examine the pairs ac , bc , cc with incoming carry zero or one. The only base relations compatible with an output in S are

$$n \in \{2a + 2, 2a + 3, 3a + 1, 3a + 2, 3a + 3\}.$$

For $n = 2a + 3$ and $n = 3a + 3$, no legal $0 \rightarrow 1$ transition exists, so c cannot be reached from the outer carry-zero state. For $n = 3a + 2$, the generic legal transitions are

$$aa : 0 \rightarrow 0 \text{ with output } 2a,$$

$$aa : 1 \rightarrow 0 \text{ with output } 2a + 1,$$

$$bc : 1 \rightarrow 1 \text{ with output } a,$$

$$cc : 0 \rightarrow 1 \text{ with output } a.$$

The only legal pair containing b is bc , and it requires carry one. The only way to enter carry one is cc , while cc is illegal with incoming carry one. Palindromicity of the pair word then forces one mirrored bc occurrence to be processed with incoming carry zero. Hence b cannot occur in the input, a contradiction. This leaves $n = 2a + 2$ and $n = 3a + 1$.

For final carry one, write $S = \{1, b, c\}$. The high outer pair is bb , bc , or cc , and its low-side output gives

outer pair	possible relation
bb	$n = 2b - 1$
bc	$n = b + c - 1$
cc	$n = 2c - 1$ or $n = 2c - b$.

If the mirrored outer occurrence has incoming carry one, its two outputs are consecutive. Output one then forces $b = 2$; apart from the small case $n = 3$, the surviving relation is $c = n - 1$, giving A_n . Output b forces $c = b + 1$ and $n = b + 2$, giving C_n .

If the mirrored outer occurrence has incoming carry zero, an interior carry drop is necessary. For $n \geq 14$, direct inspection of the six possible unordered input pairs shows that the only drop types are

$$11 \text{ with } b = 3, \quad 1b \text{ with } c = b + 2, \quad bb \text{ with } c = 2b + 1.$$

Combining these three equations with the four outer relations gives exactly twelve algebraic combinations. For completeness, Table 1 records all of them. Here a *non-leading output role* for a digit means an occurrence of that digit as the ordinary output of one of the column equations, rather than as the extra leading digit created by the final carry. In the last column, “missing input x ” means that after substitution no legal local pair containing x exists; a missing non-leading output cannot be supplied by the final carry, since that carry can supply only the leading output digit 1. These assertions follow by substituting the displayed relation into the six unordered pairs $11, 1b, 1c, bb, bc, cc$ with incoming carry 0 or 1.

Outer relation	Required drop	Consequence	Reason for exclusion / outcome
$n = 2b - 1$	$b = 3$	$n = 5$	small case, outside $n \geq 14$
$n = 2b - 1$	$c = b + 2$	$n = 2b - 1$	missing input c (and non-leading output b)
$n = 2b - 1$	$c = 2b + 1$	$c > n$	violates $c < n$
$n = b + c - 1$	$b = 3$	$n = c + 2$	non-leading output c is impossible
$n = b + c - 1$	$c = b + 2$	$n = 2b + 1$	non-leading output b is impossible
$n = b + c - 1$	$c = 2b + 1$	$n = 3b$	missing input 1 (and non-leading output b)
$n = 2c - 1$	$b = 3$	$n = 2c - 1$	missing input b and non-leading output c
$n = 2c - 1$	$c = b + 2$	$n = 2b + 3$	non-leading output b is impossible
$n = 2c - 1$	$c = 2b + 1$	$n = 4b + 1$	missing input 1 (and non-leading output b)
$n = 2c - b$	$b = 3$	$n = 2c - 3$	missing input b and non-leading output c
$n = 2c - b$	$c = b + 2$	$n = b + 4$	survives: $b = n - 4, c = n - 2$, giving B_n
$n = 2c - b$	$c = 2b + 1$	$n = 3b + 2$	missing input 1

Table 1: The twelve final-carry-one combinations when the mirrored outer occurrence has incoming carry zero.

For an explicit role check, let I_{loc} be the set of digits occurring in at least one legal unordered local input pair after substitution, and let O_{loc} be the set of digits occurring as a non-leading local output. For the ten ordered cases with $n \geq 14$ in the table, direct substitution gives, in the same order after omitting the small and order-inconsistent rows,

case	I_{loc}	O_{loc}
$(2b - 1, c = b + 2)$	$\{1, b\}$	$\{1, c\}$
$(b + c - 1, b = 3)$	$\{1, b, c\}$	$\{1, b\}$
$(b + c - 1, c = b + 2)$	$\{1, b, c\}$	$\{1, c\}$
$(b + c - 1, c = 2b + 1)$	$\{b, c\}$	$\{1, c\}$
$(2c - 1, b = 3)$	$\{1, c\}$	$\{1, b\}$
$(2c - 1, c = b + 2)$	$\{1, b, c\}$	$\{1, c\}$
$(2c - 1, c = 2b + 1)$	$\{b, c\}$	$\{1, c\}$
$(2c - b, b = 3)$	$\{1, c\}$	$\{b\}$
$(2c - b, c = b + 2)$	$\{1, b, c\}$	$\{b, c\}$
$(2c - b, c = 2b + 1)$	$\{b, c\}$	$\{b, c\}$

The final leading carry may supply output digit 1, but it cannot supply a missing input digit or a missing non-leading output b or c . Hence these role sets independently give every

exclusion in Table 1; only the ninth ordered case has all required input roles and both non-leading output roles.

Thus the only surviving combination is

$$cc : n = 2c - b, \quad 1b : c = b + 2,$$

which yields $b = n - 4$, $c = n - 2$ and therefore B_n .

Explicit witnesses are:

$$A_n : [2, 1, n - 1, n - 1, n - 1],$$

with output digit word

$$[1, 2, 1, n - 1, 1, 1],$$

$$B_n : [n - 2, 1, n - 2, n - 4, n - 2],$$

with output word

$$[1, n - 4, n - 2, n - 4, n - 2, n - 4],$$

and

$$C_n : [n - 1, 1, n - 2, n - 1, n - 1, n - 1],$$

with output word

$$[1, n - 1, 1, n - 2, n - 2, 1, n - 2].$$

For $n = 2a + 2$, one may use

$$[a, 2a + 1, a, 2a, 2a + 1, 2a + 1, 2a + 1, a],$$

and for $n = 3a + 1$,

$$[a, 2a, a, 2a + 1, a].$$

Direct column addition gives the required digit set in each case.

B Certificate inventory and reproduction protocol

The computational claims used in the paper are packaged as five finite certificate groups. The source archive also contains `README.md`, `RECORDED_OUTPUTS.txt`, `certificate_expectations.json`, `SHA256SUMS.txt`, and `verify_manifest.py`. Running the latter verifies the byte-level integrity of the distributed ancillary files.

C1. Rank at most two, $k = 6$. Source `Tn6_rank2_core_audit_fixed_v2.mpl`. In a fresh Maple session run

```
symbolic_T6_rank_explore(100, 2, 50000);
```

The decisive expected output is zero raw and zero minimal accepted systems of rank at most two; the recorded author run processed 428 live states and returned `[0, []]`.

C2. Rank three, essential support six. Source `Tn6_rank3_core_audit_fixed_v2.mpl`. Run

```
T6_rank3_core_audit(100, 50000);
```

The expected result is 25 minimal rank-three row spaces with support distribution 6, 15, 4, 0 for support sizes 3, 4, 5, 6, respectively. The corrected file performs explicit $k = 6$ dimension and feasibility self-checks before the audit.

C3. Rank three, essential support seven. Source `Tn7_primitive_rank3_core_audit.mpl`. Run

```
primitive_T7_rank3_core_audit(100, 50000);
```

The expected result is zero primitive full-support rank-three systems. The additional file `Tn8_primitive_rank3_core_audit_fixed_v2.mpl` is distributed only as an independent check; support eight is already excluded structurally by Theorem 5.3.

C4. Exact fourth column.

The exact symbolic certificate `Tn4_symbolic_exhaustivity_certificate.mpl` is run by `symbolic_T4_audit()`; it performs the symbolic BFS, reduces accepted systems by row-space containment, checks the count $60 = 7 + 53$, and compares the computed row spaces with the complete explicit list. The human family list is in `Tn4_exact_classification_proof_note.txt`. The independent direct-set source `Tn4_maple_classification_audit.txt` provides `family_audit(30)`; `witness_audit(30)`; and `exhaustive_audit(30)`; which respectively reconstruct the family union, verify every claimed set by the exact automaton, and compare with an exhaustive enumeration of all four-subsets in the selected base.

C5. Exact fifth column. In one Maple session execute, in order,

- (a) `Tn5_rank4_standalone.mpl: run_T5_rank4_new(50, 100000);`
- (b) `Tn5_rank5_new_explorer_fast.mpl: symbolic_T5_rank5_new_fast(50, 100000);`
- (c) `Tn5_rank5_analyzer_fresh.mpl: analyze_T5_rank5();`
- (d) `Tn5_rank5_remainder_exact.mpl: exact_T5_rank5_remainder();`
- (e) `Tn5_B5_congruence_compressor.mpl: compress_T5_B5();`

The decisive counts are 14, 14, 1112 in ranks 3, 4, 5, every final rank-five digit matrix has rank five, and the residue reconstruction is exact modulo 27720. The compressor finds 64 congruence signatures in total: 63 nonempty singleton-residue signatures and one empty signature of multiplicity five. Thus 1107 of the 1112 surviving affine families are arithmetically active.

Software environment and archival status

A fresh archival rerun on August 11, 2026 recorded Maple 2026.1, X86 64 WINDOWS, Jun 04 2026, Build ID 2018217 on Windows. C1–C4 completed cleanly. All mathematical C5 stages also produced the expected exact outputs, but the preserved C5 transcript contains two error messages from stale final assertions in the legacy driver, which confused 1112 surviving with 1107 active families and 64 total with 63 nonempty signatures. The supplement preserves that raw transcript and legacy driver together with an explicit postvalidation report; the distributed `C5_driver.mpl` is corrected. The current `run_archival_reruns.ps1` additionally rejects any Maple Error, line or nonempty stderr transcript, as well as missing success markers, and SHA-256 manifests fix both sources and logs.

References

- [1] K. Woods, *Presburger arithmetic, rational generating functions, and quasi-polynomials*, J. Symbolic Logic **80** (2015), 433–449.